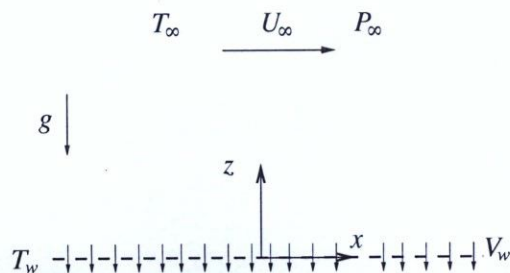


NS1

Consider the steady planar motion of a perfect liquid of density ρ , viscosity μ , thermal conductivity k and specific heat c in the vicinity of a horizontal porous wall. Far from the wall, the horizontal velocity component is $v_x = U_\infty$, the pressure is given by $p + \rho g z = P_\infty$ and the temperature is $T = T_\infty$, different from the value T_w at the wall. The liquid is sucked through the pores of the wall, resulting in a transverse velocity $v_z = -V_w$ at the wall surface. In the analysis, assume that the velocity and temperature fields are only a function of the distance to the wall z .

1. Determine the velocity components $v_x(z)$ and $v_z(z)$ and well as the pressure field $p(x, z)$.
2. Obtain the temperature field $T(z)$.
3. Calculate the streamline that crosses $x = x_0$ and $z = 0$.
4. Compute the viscous stress on the wall.
5. Determine the heat flux from the fluid to the wall.



1) $\frac{\partial}{\partial x} = 0, \frac{\partial}{\partial y} = 0, v_y = 0$
 $\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 \Rightarrow v_z = \text{constant} = -V_w$

x-MOM: $\rho v_z \frac{dv_x}{dz} = -\frac{\partial p}{\partial x} + \mu \frac{d^2 v_x}{dz^2} \Rightarrow \frac{d^2 v_x}{dz^2} + \frac{V_w}{\nu} \frac{dv_x}{dz} = 0 \Rightarrow v_x = C_1 + C_2 e^{-\frac{V_w}{\nu} z} = U_\infty (1 - e^{-\frac{V_w}{\nu} z})$

z-MOM: $0 = -\frac{\partial p}{\partial z} - \rho g \Rightarrow p + \rho g z = f(x) = P_\infty$

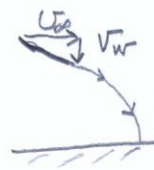
2) $-\rho c V_w \frac{dT}{dz} = \phi_r + k \frac{d^2 T}{dz^2} \rightarrow T = B_1 + B_2 e^{-\frac{V_w}{\alpha} z} + T_p$
 $\phi_r = \mu \left(\frac{dv_x}{dz} \right)^2 = \mu U_\infty^2 \frac{V_w^2}{\nu^2} e^{-2\frac{V_w}{\nu} z}$
 $T_p = \frac{Pr U_\infty^2 / c}{2(Pr-2)} e^{-2\frac{V_w}{\nu} z}$

$T(\infty) = T_\infty \rightarrow B_1 = T_\infty$

$T(0) = T_w \rightarrow B_2 = T_w - T_\infty - \frac{Pr}{2(Pr-2)} \frac{U_\infty^2}{c}$

$T = T_\infty + (T_w - T_\infty - \frac{Pr}{2(Pr-2)} \frac{U_\infty^2}{c}) e^{-\frac{V_w}{\alpha} z} + \frac{Pr}{2(Pr-2)} \frac{U_\infty^2}{c} e^{-2\frac{V_w}{\nu} z}$

3) $\frac{dx}{v_x} = \frac{dz}{v_z} \Rightarrow -V_w \int_{x_0}^x dx = \int_0^z U_\infty (1 - e^{-\frac{V_w}{\nu} z}) dz \Rightarrow x - x_0 = -\frac{U_\infty}{V_w} \left[z + \frac{\nu}{V_w} (e^{-\frac{V_w}{\nu} z} - 1) \right]$



4) $\vec{\tau} \cdot \vec{n} = \begin{bmatrix} \tau_{xx} & \tau_{xz} \\ \tau_{xz} & \tau_{zz} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = (\tau_{xz}, \tau_{zz})_{z=0}$
 $\tau_{xz} = \mu \frac{dv_x}{dz} = \mu \frac{V_w U_\infty}{\nu} e^{-\frac{V_w}{\nu} z} = \rho V_w U_\infty$
 $\tau_{zz} = 0$

5) $(\vec{q} \cdot \vec{n})_w = -k \left(\nabla T \cdot \vec{n} \right)_w = k \frac{dT}{dz} \Big|_{z=0} \Rightarrow (\vec{q} \cdot \vec{n})_w = \rho c V_w \left[T_\infty - T_w + \frac{1}{2} \frac{U_\infty^2}{c} \right]$
 $(\vec{q} \cdot \vec{n})_w = \rho V_w U_\infty \bar{e}_x$